

This section comprises six questions, each with subquestions. The overall marks are provided beside each question. These marks will be normalized according to the overall evaluation scheme. All the questions are objective answer based. Clearly mark the correct option and write the same (e.g., option a 5%-6%). Any vague response will result in a zero mark. Total marks are 38 [4+5+11+5+5+8]. The total marks will be scaled back to Nineteen (19). The formulae sheet is provided at the beginning. The exam paper runs till pg no 15. After that 3 pages are attached for rough work. The extra space around questions can also be used rough work, there are no marks for steps. Do not fail to write your name and roll no.

Name: _____ Roll No: _____

Formula Sheet

$P_0 = \frac{DIV_1}{r-g}$; where P_0 is today's price given that expected dividend at the year end Div_1 and growth rate in dividend of 'g' there onwards.

$g = \text{plowback ratio} * \text{ROE}$

Where, 'g' is the growth (in earnings), plowback ratio = 1 - payout ratio, and 'ROE' is return on equity.

Expected Returns in case of different probabilities

$$E(R_{i,t}) = \bar{R}_i = \sum_{t=1}^T P_t * R_{i,t}.$$

Variance of return for different probabilities

$$E(R_t - \bar{R}_t)^2 = \text{Variance } (\sigma_t^2) = \sum_{t=1}^T P_t (R_t - \bar{R}_t)^2$$

Expected Returns in case of the same probabilities

$$E(R_{i,t}) = \bar{R}_i = \frac{1}{T} \sum_{t=1}^T R_{i,t}.$$

Variance of return for different probabilities

$$E(R_t - \bar{R}_t)^2 = \text{Variance } (\sigma_t^2) = \sum_{t=1}^T P_t (R_t - \bar{R}_t)^2$$

Duration of a bond for T period maturity

$$\text{Duration} = 1 * \frac{PV(C_1)}{PV} + 2 * \frac{PV(C_2)}{PV} + 3 * \frac{PV(C_3)}{PV} + \dots + T * \frac{PV(C_T)}{PV}$$

CAPM: $\bar{R}_i = r_f + \beta(\bar{r}_m - r_f)$

Weighted Average Cost of Capital (WACC): $r_D * (1 - T_c) * \frac{D}{V} + r_E * \frac{E}{V}$

Question 1 [4 Marks]**A.** Compute the IRR of the project with the following cash flows.**[1 Mark]**

Project	C_0	C_1	C_2
A	-6,750	+4,500	+18,000

- (a) 85%-95%
- (b) 95%-105%;
- (c) 105%-110%;
- (d) 110%-115%;
- (e) 115%-120%;

B. Consider the three projects A, B, and C, as shown here. If the opportunity cost of capital is 10%, which of these projects have a positive NPV.

[1 Mark]

Project	C_0	C_1	C_2	C_3	C_4
A	-5,000	+2,000	+3,000	0	0
B	-1,000	0	+1,000	+2,000	+3,000
C	-5,000	+1,000	+1,000	+3,000	+5,000

- (a) A and B;
- (b) B and C;
- (c) C and A;
- (d) None of the above;

C. Compute the present value (PV) of CF= \$1 received after T=6-years, if your opportunity cost of capital is $r=12\%$. The value lies in the following interval. [1 Mark]

- (a) \$0.1-\$0.2
- (b) \$0.4-\$0.5
- (c) \$0.5-\$0.6
- (d) \$0.6-\$0.7

D. If the opportunity cost of capital is $r=9\%$, what is the appropriate interval for the present value of FV=\$374 paid in T=9th year . [1 Mark]

- (a) 160-180;
- (b) 180-200;
- (c) 200-220;
- (d) 220-240;

Question 2 [5 Marks]

The following information is given in relation to basic cash flow analysis and security valuation. Answer the questions based on the information provided.

The stock price of security A at the beginning of the year was $P_0 = \text{INR}10$. At the end of the year, the stock is selling at $P_1 = \text{INR } 9.50$. Also, at the end of the year, a dividend of $D_1 = \text{INR}1.00$ was paid.

A. Compute returns using a simple holding period return ($r = \frac{\text{Total Income}}{\text{Total Investment}}$) formula. What is the total return for the period (T=0 to T=1)? **[1 Mark]**

- (a) 1%–4%
- (b) 4%–8%
- (c) 8%–12%
- (d) 12%–16%

B. If you believe in a constant growth model of dividends ($D_1 = \text{INR } 1.00$) with a 5% growth rate (g) and cost of equity (k) at 10%; then, what should be the appropriate theoretical price (P_0) according to your analysis? **[1 Mark]**

- (a) INR 7–12
- (b) INR 13–18
- (c) INR 19–24
- (d) INR 24–29

C. In Part B, if the cost of equity is the same as the return on equity, what is the theoretical payout ratio for the firm? **[1 Mark]**

- (a) 25%–35%
- (b) 35%–45%
- (c) 45%–55%
- (d) 55%–65%

D. In Part B, if the cost of equity rises by 2% and the growth rate falls by 1%, then what is the appropriate theoretical price (assuming a constant growth model)? **[1 Mark]**

- (a) INR 0–5
- (b) INR 5–10
- (c) INR 10–15
- (d) INR 15–20

E. For another security B, everything else remains the same as security A [Dividend (D_1) = INR 1.00 and discount rate = 10%], and it is currently selling at P_0 = INR 40. What is the appropriate growth rate factored into market prices? [1 Mark]

- (a) 1%–5%
- (b) 6%–10%
- (c) 10%–15%
- (d) 15%–20%

Que 3 [11 Marks]

A. A firm has just paid a dividend of D_0 = INR 0.55 ($T = 0$). It is expecting a dividend growth rate of 10% indefinitely into the future. If the appropriate discount rate is 14%, then what is the value of the stock? [1 Mark]

- (a) INR 4–8
- (b) INR 8–12
- (c) INR 12–16
- (d) INR 16–20

B. In continuation to Part A, assume that the company decides to invest in new projects (Hint: new growth rate has to be estimated). Now, based on the overall risk of the company, the investors are demanding a 12% return from the firm. The aggregate return on all the company projects is expected to be 14%. The company expects to earn INR 2 (per share) profit in the current year and retain 50% of these profits to be reinvested in ongoing projects (the remaining will be paid as a dividend at the end of the year). What is the new price? **[2 Marks]**

- (a) INR 6–10
- (b) INR 10–14
- (c) INR 14–18
- (d) INR 19–23

C. In Part B, if all else remains the same (except the expected returns by investors), and the price is INR 30. Then, what is the estimate of expected returns? **[2 Marks]**

- (a) 4%–8%
- (b) 8%–10%
- (c) 10%–12%
- (d) 12%–14%

D. In Part A, the growth rate (10%) will continue for five years, and after that, the steady-state growth of 6% is achieved. Also, for the first five years, the payout ratio is 30%, and 50% after that. What is the fair price? **[3 Marks]**

- (a) INR 8–10
- (b) INR 10–12
- (c) INR 12–14
- (d) INR 19–23

E. In Part D, assume five years of high growth ($g = 10\%$), and after that, the growth declines linearly for next four years to reach the steady state growth of 6% ($g_6 = 9\%$, $g_7 = 8\%$, $g_8 = 7\%$, $g_9 = 6\%$). Assume that the payout ratio is constant for eight years at 30% and then changed to 50% from the end of the ninth year onward. What is the estimate of the price now? **[3 Marks]**

- (a) INR 8–10
- (b) INR 10–12
- (c) INR 12–14
- (d) INR 19–23

Question 4: The following information is given for a bond. It is a 10-year maturity bond that offers an annual coupon of 10% and a principal repayment of 1000 at maturity. The current yield to maturity of the bond is 9%. Assume annual compounding. **[5 Marks]**

A. The current price of the bond lies in the following interval. **[1 Mark]**

- (a) 1000 - 1020
- (b) 1020 - 1040
- (c) 1040 - 1060
- (d) 1060 - 1080
- (e) 1080 - 1100

B. The duration of the bond lies in the following interval. **[2 Marks]**

- (a) 5.5 - 6 years
- (b) 6 - 6.5 years
- (c) 6.5 - 7 years
- (d) 7 - 7.5 years
- (e) 7.5 - 8 years

C. If ytm is increased by 1% then the current price of the bond lies in the following interval. **[1 Mark]**

- (a) 990 - 1010
- (b) 1010 - 1030
- (c) 1030 - 1050
- (d) 1050 - 1070
- (e) 1070 - 1090

D. A bond achieves its zero coupon at its.

[1 Marks]

- (a) Maturity
- (b) Bond Coupon Rate
- (c) Durations
- (d) Bond Price
- (e) All of the above